

Review

Vectors in 3d space can be expressed as

$$\vec{v} = v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$$

Adding vectors involves adding their individual components, and scalar multiplication involves multiplying all the components by a factor.

Dot Product:

$$\begin{aligned}\vec{v} \cdot \vec{w} &= v_x w_x + v_y w_y + v_z w_z \\ &= |\vec{v}| |\vec{w}| \cos \theta \\ &= \text{proj}_{\vec{w}}(\vec{v}) \cdot |\vec{w}| \\ &= \text{proj}_{\vec{v}}(\vec{w}) \cdot |\vec{v}|\end{aligned}$$

Cross Product:

$$\vec{v} \times \vec{w} = \det \begin{pmatrix} \hat{i} & \hat{j} & \hat{k} \\ v_x & v_y & v_z \\ w_x & w_y & w_z \end{pmatrix}$$

Also remember that

$$|\vec{v} \times \vec{w}| = |\vec{v}| |\vec{w}| \sin \theta$$

Precise Definition of a Vector

(In 2-D)

A vector is a pair of quantities (v_x, v_y) whose components transform in the same way as the coordinates when the coordinate system is rotated. In other words, in a new coordinate system that is rotated θ counterclockwise from the original:

$$\begin{aligned}v_{x'} &= v_x \cos \theta + v_y \sin \theta \\ v_{y'} &= -v_x \sin \theta + v_y \cos \theta\end{aligned}$$

or

$$\begin{bmatrix} v_{x'} \\ v_{y'} \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$

(A scalar would then be a quantity that is the same in all coordinate systems.)

(In 3-D)

$$\begin{bmatrix} v_{x'} \\ v_{y'} \\ v_{z'} \end{bmatrix} = \begin{bmatrix} \cos \theta_{x'x} & \cos \theta_{x'y} & \cos \theta_{x'z} \\ \cos \theta_{y'x} & \cos \theta_{y'y} & \cos \theta_{y'z} \\ \cos \theta_{z'x} & \cos \theta_{z'y} & \cos \theta_{z'z} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

Directional Derivative

The rate of change of a scalar field when moving along a direction.

Example:

$$\begin{aligned}\frac{\partial f}{\partial q} &= a \frac{\partial f}{\partial x} + b \frac{\partial f}{\partial y} + c \frac{\partial f}{\partial z} \\ &= (a\hat{i} + b\hat{j} + c\hat{k}) \cdot \left(\frac{\partial f}{\partial x} \hat{i} + \frac{\partial f}{\partial y} \hat{j} + \frac{\partial f}{\partial z} \hat{k} \right) \\ &= \hat{e} \cdot \vec{\nabla} f\end{aligned}$$

The first term is the direction. The second term is the gradient of the scalar field f .

Gradient

The gradient is a vector in the greatest rate of change, whose magnitude is the rate of change. It can also act as a differential operator:

$$\vec{\nabla} = \frac{\partial}{\partial x} \hat{i} + \frac{\partial}{\partial y} \hat{j} + \frac{\partial}{\partial z} \hat{k}$$

In any level surface, the direction of this gradient at a point will be perpendicular to the tangent plane at that same point (i.e. it is perpendicular to any tangent line at that point). This is important for [Surface Integrals](#).

The gradient of any scalar field is a vector field.

Line Integrals and Surface Integrals

Line Integrals

A line integral of a vector field along a path is

$$\int_C \vec{v} \cdot d\vec{r}$$

where $d\vec{r}$ is the infinitesimal displacement, i.e. $dx\hat{i} + dy\hat{j} + dz\hat{k}$.

Strategies to Solve Line Integrals:

Manual

Express everything as a single variable. Use polar coordinates or parametric equations if necessary.

Fundamental Theorem of Gradients/Line Integrals

This only applies to gradients.

$\vec{\nabla} f \cdot d\vec{r} = df$, so

$$\int_C \vec{\nabla} f \cdot d\vec{r} = \int_C df = f(\vec{r}_2) - f(\vec{r}_1)$$

Stokes' Theorem and Curl

The curl of a vector is

$$\vec{\nabla} \times \vec{v} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ v_x & v_y & v_z \end{vmatrix}$$

Physically, it can be interpreted as the axis at which a vector field rotates/curles around.

Through Stokes' theorem, we can use the curl to find closed line integrals.

$$\oint_C \vec{v} \cdot d\vec{l} = \int_S (\vec{\nabla} \times \vec{v}) \cdot d\vec{A}$$

This applies regardless if planar or non-planar surfaces are considered.

Surface Integrals/Flux

Given a vector field and a level surface:

$$\int_S \vec{v} \cdot \hat{n} dA = \int_S \vec{v} \cdot d\vec{A}$$

where $d\vec{A}$ is the vector with magnitude dA and a direction perpendicular to the surface.

Closed/Gaussian Surface

$$\oint_S \vec{v} \cdot d\vec{A}$$

($d\vec{A}$ is always outward)

Generally, it is evaluated separately for each face as we are trying to measure divergence.

Alternatively, we can use the Gauss' Divergence Theorem:

$$\oint_S \vec{v} \cdot d\vec{A} = \int_V (\vec{\nabla} \cdot \vec{v}) d\tau$$

where τ is $dx dy dz$.

Intuitively, flux can be described in the analogy of fluid moving in or out a surface, where if it is multiplied by the density, it is the rate at which water exits the volume. A positive flux means it is losing fluid, negative means gaining fluid, and 0 means a steady flow.